

WEB3 DATA ANALYTICS WITH GRAPH RAG

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Abstract—The rapid development of Web3 has generated massive amounts of on-chain data, making it crucial to effectively analyze and understand the complex relationships within blockchain ecosystems. Although standard RAG techniques augment LLMs through external data retrieval, it falls short in capturing the intricate network of relationships in Web3 data. In this paper, this work introduces an innovative method that combines GraphRAG with community detection algorithms to analyze Web3 textual data. By constructing knowledge graphs from Web3-related documents and leveraging community structures, our system can better understand the semantic relationships and contextual connections in Web3 content, delivering higher-precision answers to domain-specific questions. Our experiments on real-world Web3 textual data show that our method achieves superior response accuracy and contextual understanding compared to traditional RAG approaches, especially for complex Web3 concepts and community-driven insights.

Keywords—GraphRAG; Web3; blockchain.

I. INTRODUCTION

Large Language Models (LLMs) like GPT-4 (1) have shown remarkable capabilities in tasks ranging from question answering to code generation. However, they face limitations with domain-specific knowledge and real-time information, often producing inaccurate outputs or "hallucinations" when dealing with unfamiliar patterns.

Retrieval Augmented Generation (RAG) addresses these challenges by providing LLMs with relevant external knowledge as context. While effective, RAG has limitations: it often ignores relationships between entities and are limited b "lost in the middle" problem when dealing with large text chunks.

Graph Retrieval Augmented Generation (GraphRAG) (4; 3) enhances traditional RAG by representing documents as knowledge graphs. This approach models connections between entities and provides summarized context to LLMs, improving understanding while reducing input length. However, GraphRAG's current clustering method using the Leiden algorithm focuses primarily on topological connections, potentially missing important semantic relationships between entities.

We propose an enhanced GraphRAG system for Web3 data analytics with three key innovations:

- Graph-based Knowledge Representation: A comprehensive knowledge graph capturing relationships between Web3 concepts
- Community Detection: Enhanced Leiden algorithm implementation for meaningful concept clustering
- Dual-level Search Strategy: Combined global and local search for context-aware information retrieval

Our experiments on Web3 documentation, particularly the Solana ecosystem, demonstrate significant improvements over traditional RAG approaches in accuracy and reduction of hallucinations across various query types.

II. PRIOR STUDIES

In this section, we delve into the latest advancements in Graph - augmented Large Language Models (LLMs). The integration of LLMs with graph structures that store data in natural language is a new research field. GNN-RAG(5) adopts graph neural networks to establish connectivity paths between query entities and potential answers. Subsequently, they identify the most direct route linking the query entities with the potential answer entities in order to forge the reasoning trajectories. They also fine-tune a LLaMA2-7B to answer the questions with the reasoning paths. LightRAG(4) merges a two-tiered retrieval approach with graph - augmented text indexing. They additionally reduces latency by 40

III. PRELIMINARY AND DEFINITION

Following (4) and (6), we formalize GraphRAG as a framework $\mathcal{M}$ consisting of a generation module LLM and a retrieval module $\mathcal{R}$:

$$\mathcal{M} = (LLM, \mathcal{R}(\varphi, \psi)) \quad (1)$$

where φ is the graph indexer and ψ is the graph retriever. Given a query q , the optimal answer a^* is:

$$a^* = \arg \max_{a \in A} \mathcal{M}(a|q, \mathcal{G}) \quad (2)$$

The graph database $\mathcal{G}$ is constructed from external database $\mathcal{D}$:

$$\mathcal{G} = \varphi(\mathcal{D}) = \{(h, r, t) | h \in \mathcal{V}, t \in \mathcal{V}, r \in \mathcal{E}\} \quad (3)$$

The target distribution can be decomposed as:

$$\mathcal{M}(a|q, \mathcal{G}) = \sum_{G \in \mathcal{G}} LLM(a|q, G) \cdot \psi(G|q, \mathcal{G}) \quad (4)$$

For efficiency, we approximate using the most relevant sub-graph G^* :

$$\mathcal{M}(a|q, \mathcal{G}) \approx LLM(a|q, G^*) \cdot \psi(G^*|q, \mathcal{G}) \quad (5)$$

IV. PIPELINE OF GRAPH RAG

Our implementation is based on nano-graphrag, a lightweight version of Microsoft GraphRAG that maintains the core functionality while improving efficiency.

A. Knowledge Graph Construction

The system constructs knowledge graphs from documents through the following steps:

1) Document Processing:

- Creates a dictionary with MD5 hash keys and chunked strings as values
- Implements document chunking with 128-token overlap and 1024-token maximum size
- Verifies chunk existence in key-value store database

2) Information Extraction:

- Utilizes LLM with few-shot ICL(2) for extraction
- Extracts entities with name, type (**organization, person, geo, event**), and description
- Identifies relations with source, target entities, description, and strength scores
- Performs iterative extraction until no new entities or relations are found

3) Storage and Indexing:

- Stores entities in vector database using text-embedding-3-small embeddings
- Preserves graph structure in networkX format for efficient traversal
- Concatenates entity names and descriptions for comprehensive representation

B. Community Detection

The system employs the Leiden algorithm for hierarchical community clustering, following these principles:

- Clusters knowledge graphs into multiple hierarchical levels
- Subdivides communities exceeding predefined population thresholds
- Preserves high-population communities
- Assigns community attributes to each node at various levels

The community set $\mathcal{P}$ is formally defined as:

$$\mathcal{P} = \bigcup_{i=0}^{n-1} \mathcal{P}^{(i)} \quad (6)$$

$$\mathcal{P}^{(i)} = \{p_j^{(i)} | j \in [0, m-1]\} \quad (7)$$

$$p_j^{(i)} = \{v \in \mathcal{V} | j \in \mathcal{C}(v), j \in [0, m-1]\} \quad (8)$$

where:

- $\mathcal{P}^{(i)}$ represents the i -th level community set
- $n = |\mathcal{P}|$ is the total number of levels
- $m = |\mathcal{P}^{(i)}|$ is the number of communities at level i
- $p_j^{(i)}$ denotes the set of entities in community j at level i

C. Community Report Generation

The system generates detailed community reports through:

- LLM-based analysis of entity and relation attributes within communities
- JSON-formatted reports containing:
 - Community title based on key entities
 - Structural summary
 - IMPACT severity rating
 - Detailed findings list

D. Search Strategy

The system implements two complementary search approaches:

1) *Global Search*: Designed for high-level knowledge extraction:

- Implements map-reduce paradigm for community processing
- Calculates community occurrence frequencies:
 - Maintains community cluster dictionaries
 - Computes occurrence by node frequency ratio
- Map step: Generates community relevance scores
- Reduce step: Merges descriptions and filters irrelevant information

2) *Local Search*: Optimized for fine-grained information retrieval:

- Entity retrieval:
 - Extracts [CLS] embedding from query
 - Identifies top 20 related entities
 - Applies 0.2 relevance threshold
- Community processing:
 - Identifies relevant communities from retrieved entities
 - Adds community attributes to LLM context
 - Refines results through LLM processing

3) *Search Fusion Strategy*: Final output integrates both searches via:

1) Relevance Scoring:

$$S_g = \sum_{c \in \mathcal{P}} \text{freq}(c) \cdot \text{sim}(q, c), \quad S_l = \max_{e \in \mathcal{V}} \text{sim}(q, e) \quad (9)$$

S_g : Global search score
 S_l : Local search score
 $\mathcal{P}$: Community set
 $\mathcal{V}$: Entity set
 $\text{sim}(\cdot)$: Cosine similarity

2) **Weighted Combination:**

$$\alpha \text{Co}_g + (1 - \alpha) \text{Co}_l \quad (\alpha = 0.6) \quad (10)$$

Co_g : Top-3 global communities
 Co_l : Top-5 local entities

V. EXPERIMENTS AND RESULTS

A. Data Collection and Processing

Our Web3 corpus comprises three main sources:

- 1) CryptoKGTutorial GitHub repository's Web3 documentation
- 2) Web3-related PDFs from online resources
- 3) Comprehensive Solana documentation

All documents were preprocessed to plain text format for the GraphRAG system, with emphasis on Solana's ecosystem to ensure broad coverage of Web3 concepts.

B. Query Design

We designed 50 diverse queries focusing on Solana's ecosystem, covering:

- Technical mechanisms (e.g., "Solana's proof-of-history mechanism")
- Comparative analysis (e.g., "Transaction throughput vs. traditional blockchains")
- Ecosystem development (e.g., "Solana Mobile Stack significance")

The queries test both global understanding and local knowledge retrieval capabilities.

C. Baseline Models

We compare against three RAG baselines:

- 1) **Naive RAG:** Standard k-nearest neighbors approach without structural information
- 2) **Light RAG:** Dual-level retrieval with graph-enhanced text indexing
- 3) **Hybrid RAG:** Combined traditional RAG and lightweight graph structures

D. Evaluation Metrics

Using GPT-4o as an automated judge, we evaluate responses across five dimensions:

- 1) **Accuracy:** Factual correctness and alignment with Web3 documentation
- 2) **Comprehensiveness:** Coverage of relevant aspects from high-level to technical details
- 3) **Diversity:** Integration of information from various Web3 sources

- 4) **Empowerment:** Clarity and actionability of explanations
- 5) **Hallucination:** Reliability and verifiability of information

The evaluation process ensures:

- Consistent criteria across comparisons
- Detailed reasoning with supporting examples
- Balance of technical accuracy and practical utility

E. Experimental Results

Table I shows the head-to-head comparison win rates (%) between our approach and baseline models across different evaluation dimensions.

The phenomenon corroborates the distinguished function of our GraphRAG method across all evaluation dimensions. Most notably, when compared to Naive RAG, our system achieves perfect scores (100%) across all metrics, indicating a significant improvement over the baseline approach. Against Light RAG, our system shows particularly strong performance in comprehensiveness (78%) and accuracy (72%), though the advantages in diversity (54%) and hallucination prevention (53.1%) are more modest. Compared to hybrid RAG, our approach maintains consistent advantages, with the strongest improvements in accuracy (66.0%) and hallucination prevention (64.0%).

Our findings confirm the efficacy of the community-aware approach, particularly in cases requiring deep comprehension and reliable information retrieval. The perfect performance against Naive RAG underscores the importance of incorporating graph structure and community detection in Web3 knowledge processing. The more moderate improvements against Light RAG and Hybrid RAG suggest that while these advanced baselines capture some of the benefits of graph-based approaches, our full GraphRAG system still provides substantial advantages, especially in terms of accuracy and comprehensiveness.

F. Ablation Studies

We managed ablation studies to assess the attempt of each component by selectively removing or modifying critical parts of the system:

Table II shows the ablation study results, comparing performance metrics (%) with the full GraphRAG system.

The ablation results highlight several key findings:

- **Community Detection:** Removing community awareness leads to a significant drop in performance (average 32.7% decrease), particularly affecting diversity (37.6% drop) and hallucination prevention (34.2% drop).
- **Graph Structure:** The graph-based knowledge representation proves fundamental, with its removal resulting in the largest performance degradation (average 46.9% decrease).

Table I
WIN RATES OF GRAPHRAG AGAINST BASELINE MODELS (%)

Method	Metric	GraphRAG vs Naive RAG	GraphRAG vs Light RAG	GraphRAG vs Hybrid RAG
Web3 Dataset				
	Accuracy	100%	72%	66.0%
	Comprehensiveness	100%	78%	58.0%
	Diversity	100%	54%	49.0%
	Empowerment	100%	62%	52.0%
	Hallucination	100%	53.1%	64.0%

Table II
ABLATION STUDY RESULTS SHOWING PERFORMANCE OVERALL METRICS (%) COMPARED TO FULL GRAPHRAG SYSTEM.

Variant	Accuracy	Comprehensiveness	Diversity	Empowerment	Hallucination
Full GraphRAG	100	100	100	100	100
w/o Community Detection	67.3	62.4	62.4	70.5	65.8
w/o Graph Structure	53.1	52.1	48.7	55.2	51.3
w/o Dual-level Search	65.7	61.4	57.2	63.8	59.1

- **Dual-level Search:** The absence of dual-level search strategy shows a substantial impact (average 38.6% decrease), particularly affecting comprehensiveness (38.6% drop) and accuracy (34.3% drop).

These results validate our design choices and demonstrate the synergistic effect of combining community detection, graph-based knowledge representation, and dual-level search in the Web3 domain. Each component contributes significantly to the overall performance, with the graph structure being the most critical element.

VI. CONCLUSION AND FUTURE WORK

Our research demonstrates that GraphRAG's community-aware approach significantly improves the quality of Web3 data analytics. The integration of graph-based knowledge representation with community detection provides a more nuanced understanding of complex blockchain ecosystems. Key contributions include:

- 1) A novel approach to structuring Web3 knowledge using community-aware graphs
- 2) Significant improvements in response accuracy and comprehensiveness
- 3) Reduced hallucination rates through better context validation

Future work will focus on:

- Extending the system to handle real-time updates to Web3 knowledge
- Implementing cross-chain community detection for better inter-blockchain analysis
- Developing specialized community detection algorithms for different types of Web3 data
- Exploring dynamic community restructuring based on query patterns

Our results suggest that community-aware approaches could be valuable not just for Web3 data analytics, but for any domain with complex, interconnected knowledge structures. We believe this work provides a foundation for

more sophisticated knowledge management systems in the rapidly evolving blockchain space.

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